
A Conditional Coupling–Diversity–Precision Bound for TTRL-Based LLM Evaluation

Anonymous Authors

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Abstract

We study a conditional coupling–diversity–precision bound for a specified TTRL observation model. The primary theoretical estimand is the coefficient of variation of a regular estimator of a scalar coupling parameter γ under independent Bernoulli observations with logistic success probability. Under this model, the Cramer–Rao inequality yields a precision floor. A positive product bound additionally requires a compact parameter space and an assumed positive lower bound on strategy entropy; it is not a universal impossibility theorem for arbitrary evaluators or protocols. Empirically, all eleven condition rows are reconstructed from the persisted per-repetition archives under a single estimator (per-repetition mean γ ; full-archive CV, with five-seed subsample ranges), so the consolidated paper supports complete γ /CV condition-level verification on the reconstruction. The observed grid contains no point in the region $\{\gamma < 0.2, \text{CV} < 0.3\}$; this emptiness is robust to threshold perturbations under the full-archive CV estimator (both low- γ rows have full-archive $\text{CV} \geq 2.3$); five-seed subsample CVs can be lower, so subset-level emptiness is estimator-dependent. Chatbot Arena correlations are observational external associations rather than validation of the TTRL bound. The contribution is therefore a checkable conditional model, an explicit theory-to-estimator map, and a list of robustness tests required before broader generalization.

1 Introduction

When large language models serve as evaluators in adaptive systems, their biases propagate into agent strategy distributions through Test-Time Reinforcement Learning (TTRL). This evaluator preference coupling creates a measured tradeoff: the source-study five-row table spans $\gamma \in [0.033, 1.038]$ and $\text{CV}(N=5) \in [0.157, 2.42]$, while the archive-based reconstruction (Table 3) spans $\gamma \in [0.033, 1.176]$ and full-archive $\text{CV} \in [0.196, 5.477]$: stronger evaluator signal produces more stable measurements (Ablation max: $\gamma = 1.038$, $\text{CV}(N=5) = 0.157$) but introduces more bias (lower strategy diversity); weaker evaluator signal preserves strategy diversity (DS Self-Eval: $H = 0.999$) but produces noisy, unreliable measurements ($\text{CV}(N=5) = 2.42$, a $15.4\times$ inflation).

We ask when a coupling–diversity–precision product has a positive lower bound. Proposition 3 answers this only for the stated logistic observation model and entropy-floor assumption. The empirical grid is used to illustrate, not prove, that conditional result.

Our contributions are:

1. A conditional precision bound with all observation-model and regularity assumptions stated explicitly.

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2. A theory-to-estimator map distinguishing measured quantities from modeling assumptions and empirical heuristics.
 3. A scoped empirical audit of the five condition rows reproduced in this manuscript plus observational Chatbot Arena associations.
 4. **A standardized protocol** for within-condition verification with convergence dynamics analysis.
 5. A robustness agenda for alternative metrics, thresholds, binning, exclusions, and leave-one-condition-out analysis.

2 Framework

2.1 The EPC Framework

An executor agent maintains a weight vector $\mathbf{w} \in \Delta^{K-1}$ over K strategies. At each round t :

1. Sample strategy $s_t \sim \mathbf{w}^{(t)}$
2. Generate response using strategy s_t
3. Receive evaluator judgment $y_t \in \{0, 1\}$
4. Update weights: $w_{s_t}^{(t+1)} = w_{s_t}^{(t)} \cdot \exp(\alpha_{y_t})$ (multiplicative reweighting)

2.2 Core Definitions

Definition 1 (Evaluator Coupling). *The coupling coefficient*

$$\gamma_{A \rightarrow B} = \frac{\|\mathbf{w}_{A \rightarrow B} - \mathbf{w}_B\|_2}{\|\mathbf{w}_B\|_2}$$

measures how much evaluator A's preferences distort executor B's strategy distribution. $\gamma \approx 0$ indicates no influence; $\gamma > 1$ indicates influence exceeding natural strategy concentration.

Definition 2 (Strategy Entropy). *$H = -\frac{1}{\log K} \sum_{i=1}^K w_i \log w_i$ is the normalized Shannon entropy of the strategy distribution, measuring diversity. $H = 1$ indicates uniform distribution; $H = 0$ indicates deterministic strategy selection.*

Definition 3 (Measurement Noise). *$\text{CV}(N) = \text{std}(\hat{\gamma}_N) / \mathbb{E}[\hat{\gamma}_N]$ is the coefficient of variation of the coupling estimate at sample size N , measuring the stability of coupling measurements; *std* is the sample standard deviation (Bessel-corrected denominator).*

3 Conditional Observation Model and Bounds

3.1 Observation Model and Regularity Conditions

Let $Y_1, \dots, Y_N$ be conditionally independent Bernoulli observations with

$$P_\gamma(Y_i = 1) = s(\gamma) = \{1 + \exp(-\gamma)\}^{-1}, \quad \gamma \in \Theta = [\gamma_{\min}, \gamma_{\max}],$$

where $0 < \gamma_{\min} < \gamma_{\max} < \infty$. The sampling unit is one independent seed-level evaluator outcome; repeated rounds within a seed do not satisfy this independence assumption and must first be aggregated or modeled hierarchically. We assume an unbiased regular estimator $\hat{\gamma}_N$, differentiability of the likelihood, parameter-independent support, finite Fisher information, and interchangeability of differentiation and integration. These assumptions describe a stylized scalar logistic model, not the raw multi-strategy TTRL trace.

Proposition 1 (Conditional precision floor). *Under the model and regularity conditions above,*

$$\text{CV}(N) = \frac{\sqrt{\text{Var}(\hat{\gamma}_N)}}{\gamma} \geq \frac{1}{\gamma \sqrt{Ns(\gamma)(1-s(\gamma))}}.$$

Proof. For one observation, $\ell_i(\gamma) = Y_i \log s(\gamma) + (1 - Y_i) \log(1 - s(\gamma))$. Since $s'(\gamma) = s(\gamma)(1 - s(\gamma))$,

$$\frac{\partial \ell_i}{\partial \gamma} = Y_i - s(\gamma), \quad I_1(\gamma) = \mathbb{E}_\gamma[(Y_i - s(\gamma))^2] = s(\gamma)(1 - s(\gamma)).$$

Conditional independence gives $I_N(\gamma) = NI_1(\gamma)$. The Cramer–Rao inequality gives $\text{Var}(\hat{\gamma}_N) \geq I_N(\gamma)^{-1}$; dividing its square root by $\gamma > 0$ yields the result. $\square$

Proposition 2 (Variance of the empirical distance proxy). *Let w_t follow the multiplicative update $\log w_{t+1,i} = \log w_{t,i} + Z_{t,i}$, where Z_t are independent across rounds with $\mathbb{E}[Z_i] = \mu_i$ and $\text{Var}(Z_i) = \sigma_i^2$ (the Bernoulli preference draws of the TTRL protocol), and let $\hat{\gamma}_N = \|w_A - w_B\|_2 / \|w_B\|_2$ be the empirical coupling proxy at horizon N with normalized weights. Then, to first order in the Delta method,*

$$\mathbb{E}[\hat{\gamma}_N] = g(\mu) + O(N^{-1}), \quad \text{Var}(\hat{\gamma}_N) = \frac{1}{N} \sum_j \left(\frac{\partial g}{\partial \mu_j} \right)^2 \sigma_j^2 + O(N^{-2}), \quad (1)$$

where g is the normalized-distance map and the first-order term scales the per-round preference variance by the sensitivity of the proxy to each strategy’s drift. In the independent-domain simulation described below ($K=11$, win $\times 1.08$ / lose $\times 0.96$, $\mathbb{P}(\text{win}) = s(\gamma)$), the realized $\text{Var}(\hat{\gamma}_N)$ grows approximately linearly in N ($0.00034 \rightarrow 0.0022$ as $N:5 \rightarrow 30$), because the proxy distance itself grows with the trajectory length and its per-round variance does not average out; meanwhile the Bernoulli Cramér–Rao floor $1/(Ns(\gamma)(1-s(\gamma)))$ decays as $1/N$. Hence

$$\frac{\text{Var}(\hat{\gamma}_N)}{1/(Ns(\gamma)(1-s(\gamma)))} \sim N^2 \cdot c(\gamma), \quad c(\gamma) \approx 1.3 \times 10^{-5} \cdot 30^2 / 0.17 \approx 0.07 \quad (\text{at } \gamma=1, N=30), \quad (2)$$

so the conditional CR floor is severely non-tight for the trajectory proxy: the bound is far below the actual variance and the gap widens with N .

Proof sketch (Delta method). Write $\log w_t = \log w_0 + \sum_{s=1}^t Z_s$; the CLT for the random walk gives $\log w_N = \log w_0 + N\mu + \sqrt{N}\mathcal{N}(0, \Sigma)$ asymptotically, where $\Sigma = \text{diag}(\sigma_j^2)$ under independent strategies. After normalization, $\hat{\gamma}_N = g(\bar{\mu}_N)$ with $\bar{\mu}_N \rightarrow \mu$; a first-order Taylor expansion about μ yields the mean and variance expressions above. The N dependence of the realized variance is set by the proxy map g : when the distance grows with trajectory length (independent-domain training, simulated), the leading variance term carries an extra factor of N relative to a mean estimator, and the ratio to the $1/N$ Bernoulli floor grows as N^2 . When instead the sampling units are the Bernoulli preferences themselves (e.g., a direct logistic estimate of γ), the standard CR analysis of Proposition 1 applies and is tight in the regular-estimator regime. The empirical E4 archive ($n=30$, $\text{Var}(\hat{\gamma})=0.158$, floor 0.176, ratio 0.90) sits between these extremes, indicating that the live coupling process includes domain-transfer structure that the independent-domain simulation omits. $\square$

Estimator definition and bias. The proxy $\hat{\gamma}_N = \|w_A - w_B\|_2 / \|w_B\|_2$ is a distance between normalized weight trajectories, not a logistic estimate of γ ; it is therefore not an unbiased estimator of γ in general. In the independent-domain simulation, $\mathbb{E}[\hat{\gamma}_N]$ is governed by the trajectory length N (its realized range is ≈ 0.08 – 0.20 across $\gamma \in [0.05, 3.0]$ at $N \in \{5, 10, 30\}$) rather than by the

true coupling strength, so the proxy carries almost no information about γ under independent training. Consequently the standard Cramér–Rao analysis of Proposition 1—which assumes an unbiased regular estimator of γ —does not apply to $\hat{\gamma}_N$ in that regime; Proposition 2 reports the variance of the proxy as an empirical quantity and shows that the Bernoulli floor is both non-tight and inapplicable to it. When the data-generating process includes domain transfer (as in the live E4 archive), the seed-level variance of the proxy approaches the Bernoulli floor ($\text{Var}/\text{floor} \approx 0.90$ at $n=30$), indicating that the proxy becomes informative about γ and that the floor is approximately tight for the seed-level sampling units. Readers should therefore interpret Proposition 2 as a caveat about the proxy, not as a CR-efficiency claim for it.

Empirical coverage of the precision floor and MLE closeness. We probe the proposition with a deterministic simulation and with the eleven persisted archives (script `evidence/scripts/p2_mle_and_theorem_coverage_20260808.py`; output `analyses/p2_mle_theorem_20260808.json`). **Coverage:** simulating $Y \sim \text{Bernoulli}(s(\gamma))$ and estimating $\hat{\gamma}$ by the logit of the sample proportion (5,000 runs per cell), the empirical CV fails to meet the lower bound $1/(\gamma\sqrt{Ns(\gamma)(1-s(\gamma))})$ in 10 of the 21 probed cells: $(\gamma, N) = (0.05, 30)$ (6.794 vs. bound 7.305), $(0.1, 10)$ (6.015 vs. 6.333), $(0.25, 10)$ (2.422 vs. 2.550), $(1.0, 5)$ (1.007 vs. 1.009), $(1.0, 10)$ (0.683 vs. 0.713), $(2.0, 5)$ (0.426 vs. 0.690), $(2.0, 10)$ (0.326 vs. 0.488), $(3.0, 5)$ (0.238 vs. 0.701), $(3.0, 10)$ (0.180 vs. 0.496), and $(3.0, 30)$ (0.186 vs. 0.286); the remaining 11 cells meet the bound. The pattern is uniform: the logit estimator has first-order finite-sample bias $\mathbb{E}[\hat{\gamma}] - \gamma \approx (1 - 2s)/(2Ns(1 - s))$, which inflates with $1/(Ns(1 - s))$; failures concentrate where N is small ($N=5-10$ at $\gamma \geq 0.1$) or where $s(\gamma)$ approaches the boundary ($\gamma=3$, $s=0.953$, fails even at $N=30$). The estimator is therefore not the unbiased regular estimator assumed by Cramér–Rao in exactly the cells where the bound matters most. The proposition is a statement about the regular-estimator regime, not a universal inequality—matching the conditional framing of the observation model above and of the product bound below. **MLE closeness:** fitting the Bernoulli model $Y_i = \mathbf{1}[\gamma_i > 0]$ to the eleven archives yields a finite logit-MLE in only two conditions (DS Self-Eval: $\hat{\gamma}_{\text{MLE}} = -3.37$, positive-coupling rate 0.033; DS×Qwen: -1.61 , rate 0.167); in the other nine conditions every repetition is positively coupled (rate 1.0), so the Bernoulli likelihood carries no information and the MLE is undefined. Where it is defined, the MLE has the same condition ordering as the pipeline proxy but takes negative values, violating the model’s $\gamma > 0$ assumption. This confirms the Table 1 caveat that γ is not estimated from Bernoulli likelihood in the current pipeline: the seed-level coupling indicators are far too coarse to identify the logistic parameter, and the proposition’s regularity conditions are correspondingly not satisfied at the self-evaluation conditions that dominate the low- γ regime.

3.2 Additional Assumptions Needed for a Product Bound

The observed negative γ – H association does not by itself imply a deterministic entropy law. We therefore state the missing assumption explicitly:

Entropy-floor assumption. On the compact parameter region Θ and for a fixed TTRL update rule, normalized strategy entropy satisfies $H(\gamma) \geq h_{\min} > 0$.

Proposition 3 (Conditional product bound). *Under the logistic observation model, regularity conditions, and entropy-floor assumption,*

$$\gamma H(\gamma) \text{CV}(N) \geq \frac{2h_{\min}}{\sqrt{N}} > 0.$$

Table 1: Theory-to-measurement map and assumption audit.

Symbol	Theoretical role	Empirical construction / limitation
γ	Scalar logistic parameter	Normalized distance between evaluator-conditioned and baseline strategy weights; not estimated from Bernoulli likelihood in the current pipeline
H	Entropy function with floor $h_{\min}$	Normalized entropy of final strategy weights; floor is assumed, not proved
$\text{CV}(N)$	Relative SD of regular $\hat{\gamma}_N$	Seed-level SD divided by seed-level mean; round dependence may violate independence
N	Independent sampling units	Number of seeds, not number of rounds or API calls
σ_w	Weight-coordinate dispersion	Empirical summary used in a heuristic continuous-weight expression
$\mathbf{w}_0$	Fixed initial weight vector	Protocol input; changes across protocols alter the estimand

Proof. Proposition 1 gives

$$\gamma H \text{CV} \geq \frac{H(\gamma)}{\sqrt{Ns(\gamma)(1-s(\gamma))}}.$$

Because $s(\gamma)(1-s(\gamma)) \leq 1/4$ and $H(\gamma) \geq h_{\min}$, the right-hand side is at least $2h_{\min}/\sqrt{N}$. $\square$

This proposition is conditional rather than universal. It fails to apply when observations are dependent, the estimator is biased/nonregular, γ approaches zero, or entropy has no positive floor. The empirical quantity $\sigma_w/(\|\mathbf{w}_0\|_2\sqrt{N})$ is retained only as a data-dependent heuristic for continuous weights; without a sampling theorem connecting σ_w to Fisher information, it is not labeled a tight lower bound.

4 Empirical Validation

Statistical Note: Multiple Comparisons

We apply family-wise Bonferroni correction to control the family-wise error rate across multi-test families in this paper. For a family of k statistical tests answering a single empirical question, the corrected threshold is $\alpha' = 0.05/k$. Significance stars below are reported against the corrected levels:

- *: $p < 0.05/k$
- **: $p < 0.01/k$
- ***: $p < 0.001/k$

The three multi-test families used in the empirical sections are:

- γ - H family ($k = 2$, $\alpha' = 0.025$): body 5-row $r = -0.989$; the abstract 11-LLM $r = -0.868$ is a source-study value that cannot be recomputed locally because the archive does not persist per-repetition H .
- Chatbot Arena correlation family ($k = 2$, $\alpha' = 0.025$): entropy-decisiveness $r(H, \text{dec}) = -0.895$; CV-decisiveness $r(\text{CV}, \text{dec}) = -0.075$.

Table 2: Five condition rows as reported in the source study (CV($N=5$) on the study’s five-seed sets). The archive-based reconstruction of all eleven rows appears in Table 3 with per-repetition estimators; the two tables are not one-to-one (see mapping caveat below); this table therefore cannot support an all-11-condition audit.

Condition	Evaluator	γ	H	CV($N=5$)
DS Self-Eval	DeepSeek	0.033	0.999	2.42
DS $\times$ Qwen	DeepSeek	0.187	0.988	1.025
Qwen $\times$ DS	Qwen	0.766	0.931	0.157
Ablation mid	Mixed	0.560	0.954	0.184
Ablation max	Mixed	1.038	0.913	0.157

The remaining six source-study conditions are represented by seven reconstruction rows in Table 3 (the mapping is not one-to-one; s

- Convergence-dynamics family ($k = 2$, $\alpha' = 0.025$): half-life- α_{eval} $r = 0.76$; ΔH - α_{eval} $r = 0.95$.

Stars next to each p -value in the text below denote the corrected significance level; “NS” denotes non-significance at the corrected threshold.

4.1 Cross-Condition Validation (11 Conditions)

We validate across 11 evaluator-agent conditions spanning the source study’s evaluator models (GPT-4o, DeepSeek-V3, Qwen-3.7, Claude-3.5) with three executor models and two experimental protocols; the archive reconstruction covers three of the four evaluator models (Claude-3.5 has no persisted row).

For the five reproduced rows, the Pearson correlation is $r = -0.989$; with $n = 5$, this estimate is highly sensitive to individual rows and is reported descriptively. A leave-one-condition-out interval and robust/permutation correlations require the missing condition-level data. The smallest observed product in the five-row table is 0.080 (DS Self-Eval: $0.033 \times 0.999 \times 2.42$); that fact does not identify the entropy floor or validate Proposition 3 because the empirical γ estimator does not follow the stipulated Bernoulli likelihood.

Archive-based eleven-condition reconstruction. Using the persisted per-repetition archives, we reconstruct all eleven conditions with a single estimator (per-repetition mean γ ; CV over the full archive; five-seed subsample range): Table 3. The empty favorable region $\{\gamma < 0.2, \text{CV} < 0.3\}$ remains empty on the reconstruction: the two low- γ rows (DS Self-Eval 0.033, DS $\times$ Qwen 0.187) have full-archive $\text{CV} \geq 2.3$, far above any reasonable CV cut.

Mapping caveat. Table 2 and Table 3 are not one-to-one: the source-study table reports CV($N=5$) on specific five-seed sets and includes Ablation mid, whose archive is not persisted; the reconstruction reports per-repetition mean γ and full-archive CV with five-seed subsample ranges. Differences (Qwen $\times$ DS 0.766 vs. 0.9866; Ablation max CV 0.157 vs. 0.366; DS Self-Eval CV 2.42 vs. 5.477) follow from these estimator and seed-set differences. The repeated 2.236 ($= \sqrt{5}$) in the CV5 ranges is not a placeholder: for DS Self-Eval, 29 of 30 archived repetitions are zero-coupling, so any five-seed subset containing the single non-zero value has mean $x/5$ and sd $x/\sqrt{5}$, giving $\text{CV} = (x/\sqrt{5})/(x/5) = \sqrt{5}$ (using the sample standard deviation with Bessel-corrected denominator); all-zero subsets have undefined CV and are excluded. The DS Self-Eval row’s low- γ /high-CV position rests on a single non-zero repetition (29/30 zero-coupling); per-seed γ values are provided in the evidence package, and this row should be treated as fragile when anchoring the frontier.

Table 3: **Archive-based eleven-condition reconstruction.** γ = per-repetition mean; CV = sd/mean over the full archive; CV5 range = five-seed subsample distribution.

Condition	Archive	n	γ	CV	CV5 range
DS Self-Eval	mm_epc_multi_seed_ds_final.json	30	0.0332	5.477	[2.236, 2.236]
DS x Qwen (qwen eval)	mm_epc_multi_seed_final.json	30	0.1865	2.306	[0.587, 2.236]
Qwen x DS (official A)	official_A_qwen_eval_ds_exec.json	5	0.9866	0.241	[0.241, 0.241]
DS eval / Qwen exec (official B)	official_B_deepseek_eval_qwen_exec.json	5	0.8929	0.336	[0.336, 0.336]
GPT4o eval / DS exec len500 (official C)	official_C_gpt4o_eval_ds_exec_len500.json	5	0.7949	0.389	[0.389, 0.389]
GPT4o eval / Qwen exec (official D)	official_D_gpt4o_eval_qwen_exec.json	5	0.7238	0.357	[0.357, 0.357]
Ablation max	mm_epc_ablation_max.json	10	1.0384	0.366	[0.145, 0.511]
Ablation no-s0	mm_epc_ablation_no_s0.json	5	0.9790	0.400	[0.400, 0.400]
Qwen37	mm_epc_qwen37_final.json	8	1.0591	0.260	[0.143, 0.355]
GPT4o replication (OLD)	mm_epc_gpt4o_replication_OLD.json	8	1.1763	0.324	[0.208, 0.400]
DS self-eval r30	mm_epc_ds_selfeval_r30.json	10	0.9360	0.196	[0.066, 0.208]

The source-study $CV(N=5)=2.42$ for this row uses a different five-seed set and/or SD denominator than the reconstruction’s 2.236.

Three robustness checks are computable from the persisted archives. (i) **Threshold robustness (full-archive CV estimator).** For every γ -cut in $[0.1, 0.5]$ and CV-cut in $[0.15, 1.0]$, the region $\{\gamma < \gamma_{\text{cut}}, CV < CV_{\text{cut}}\}$ is empty for the archive reconstruction, because the two low- γ rows have full-archive $CV \geq 2.3$. Subset-level emptiness is not claimed: five-seed subsample CVs can be as low as 0.587 for DS $\times$ Qwen (see (ii)), so a five-seed analysis could place that row inside a CV-cut of 0.6. (ii) **Seed-subset dependence of $CV(N=5)$.** Five-seed subsamples from the persisted 30-repetition archives yield CV in $[0.587, 2.236]$ (median 2.236) for DS $\times$ Qwen versus the reported 1.025, and the 252 five-seed subsets of Ablation max span $[0.145, 0.511]$ (median 0.355) versus 0.157; we therefore recommend reporting full-archive CV alongside $CV(N=5)$. (iii) **Proxy-JSD alignment.** The empirical γ proxy correlates near-perfectly with the Jensen–Shannon distance between evaluator-conditioned and baseline strategy weights (Pearson $r = 0.969$, Spearman 0.998, $n = 152$; leave-one-source-out Pearson range $[0.968, 0.983]$), quantifying that the measured γ behaves as a distributional-distance proxy rather than the theorem’s Bernoulli-likelihood estimator; the theory-to-estimator gap in Table 1 is therefore quantitative, not merely qualitative. (iv) **CV-estimator sensitivity.** Table 4 compares the full-archive CV (the estimator used for the reconstruction and the threshold-robustness checks) with the distribution of CV computed on five-seed subsamples of the same persisted archives. For the conditions with $n=10$ –30 repetitions, the five-seed subsample median tracks the full-archive CV (Ablation max: 0.355 vs. 0.366; DS self-eval r30: 0.197 vs. 0.196; Qwen37: 0.250 vs. 0.260). The two $n=30$ self-evaluation archives are dominated by zero-coupling repetitions: five-seed subsamples either consist entirely of zero-coupling repetitions (for which CV is undefined) or collapse to the analytic $\sqrt{5}$ value derived in the mapping caveat above (DS Self-Eval: 2.236 for every non-degenerate subsample). We therefore use the full-archive CV for the empty-region analysis: it is the single estimator that is defined for all eleven conditions, maximizes sample efficiency, and is stable against the five-seed subsample distribution wherever that distribution is non-degenerate. The favorable regime $\{\gamma < 0.2, CV < 0.3\}$ remains empty under either estimator.

4.2 External Validation: Chatbot Arena (20 Models)

We analyze 55,000 publicly available human preference judgments from Chatbot Arena across 20 models. The entropy–decisiveness correlation replicates: $r(H, \text{dec}) = -0.895$ ($p < 0.001^{**}$)

Table 4: **CV-estimator sensitivity.** Full-archive CV vs. five-seed subsample CV (median with [min,max] range over non-degenerate five-seed subsamples). Script: `evidence/scripts/verify_p2_cv_sensitivity_20260808.py`; output `analyses/p2_cv_sensitivity_20260808.json`.

Condition	n	Full-archive CV	CV5 subsample median	CV5 subsample range
DS Self-Eval	30	5.477	2.236	[2.236, 2.236]
DS x Qwen (qwen eval)	30	2.306	2.236	[0.587, 2.236]
Qwen x DS (official A)	5	0.241	0.241	[0.241, 0.241]
DS eval / Qwen exec (official B)	5	0.336	0.336	[0.336, 0.336]
GPT4o eval / DS exec (official C)	5	0.389	0.389	[0.389, 0.389]
GPT4o eval / Qwen exec (official D)	5	0.357	0.357	[0.357, 0.357]
Ablation max	10	0.366	0.355	[0.145, 0.511]
Ablation no-s0	5	0.400	0.400	[0.400, 0.400]
Qwen37	8	0.260	0.250	[0.143, 0.328]
GPT4o replication (OLD)	8	0.324	0.325	[0.208, 0.417]
DS self-eval r30	10	0.196	0.197	[0.066, 0.286]

Table 5: **The evaluation frontier.** The favorable regime $\{\gamma < 0.2, \text{CV} < 0.3\}$ is empirically empty.

Regime	Example	γ	CV($N=5$)
Low- γ , high-CV	DS Self-Eval	0.033	2.42
High- γ , low-CV	Ablation max	1.038	0.157
Low- γ , low-CV	<i>None</i>	—	—

under Bonferroni correction for the Chatbot Arena family with $k = 2$, $\alpha' = 0.025$). Models with clear pairwise dominance (GPT-4o, GPT-4-turbo) show lower outcome entropy than models with ambiguous comparisons.

The CV-decisiveness correlation is $r(\text{CV}, \text{dec}) = -0.075$ with 95

4.3 The Empirical Frontier

The frontier is bimodal: conditions cluster at either the low- γ , high-CV corner or the high- γ , low-CV corner. No condition occupies $\{\gamma < 0.2, \text{CV} < 0.3\}$. At the reported point estimates this emptiness is threshold-robust within the five fully reported rows (both low- γ rows have $\text{CV}(N=5) \geq 1.025$) and persists on the archive-based eleven-condition reconstruction (Table 3), where the two low- γ rows have full-archive $\text{CV} \geq 2.3$.

5 Within-Condition Verification

5.1 Protocol Design

We provide a standardized protocol for within-condition verification of the bias-reliability tradeoff:

- Single LLM serves as both executor and evaluator
- Independent variable: evaluator feedback strength $\alpha_{\text{eval}} \in \{0, 0.5, 1.0\}$
- Configuration: $N \geq 20$ seeds, $T \geq 30$ rounds, 6 tasks, 11 strategies

- Predictions: (P1) $\text{corr}(\alpha_{\text{eval}}, \gamma_{\text{meas}}) > 0$; (P2) $\text{corr}(\alpha_{\text{eval}}, H) < 0$; (P3) $\text{corr}(\gamma_{\text{meas}}, \text{CV}) < 0$

5.2 Convergence Dynamics

TTRL simulation ($N=50$ seeds, $T=30$ rounds) reveals three key findings:

Finding 1: Gradual, monotonic collapse. Strategy entropy $H(t)$ decreases smoothly from $H(0) \approx 0.9999$ to a steady-state value. No phase transition, no critical α_{eval} . Half-life: 14–19 rounds, largely independent of α_{eval} ($r = 0.76$, $p = 0.13$, NS under Bonferroni correction for the convergence-dynamics family with $k = 2$, $\alpha' = 0.025$).

Finding 2: Evaluator strength controls magnitude. Total entropy drop $\Delta H = H(0) - H(30)$ scales with α_{eval} : from $\Delta H = 0.0043$ at $\alpha_{\text{eval}} = 0$ to $\Delta H = 0.0128$ at $\alpha_{\text{eval}} = 2.0$ ($r = 0.95$, $p = 0.012^*$ under Bonferroni correction for the convergence-dynamics family with $k = 2$, $\alpha' = 0.025$).

Finding 3: Evaluator pressure increases heterogeneity. Across-seed variance in final entropy *increases* with α_{eval} : σ_{final} from 0.0022 to 0.0039. Stronger evaluator preferences produce greater heterogeneity in convergence outcomes.

Power analysis. The half-life of 14–19 rounds implies $T \geq 30$ for $> 85\%$ convergence. At $T=30$, the entropy difference between $\alpha_{\text{eval}} = 0$ and 1.0 has Cohen’s $d \approx 1.2$, providing $> 80\%$ power at $N=20$ seeds. Recommended: 3–5 α_{eval} levels, 20–30 seeds per level, 30 rounds per seed ($\approx 5,400$ API calls, $\approx \$0.50$).

6 GPT-4o Version Drift

A single two-snapshot observation motivates longitudinal monitoring: GPT-4o coupling differs between two API snapshots one month apart ($\Delta\gamma = -1.176$; May 2026: $\gamma \approx 1.176$, $N=8$; June 2026: $\gamma = 0.000$, $N=8$, fully zero-coupling). The June snapshot’s 100% zero-coupling needs seed-level corroboration; we treat this as a monitoring prompt, not an established drift law. Both snapshots used identical prompts, identical seeds, identical evaluation metrics; the only difference was the model snapshot. The claim that coupling measurements can invert within 4 weeks (magnitude $\Delta\gamma = 1.176$ at $N=8$) is refuted if a re-measurement in July 2026 under identical conditions yields γ within ± 0.2 of either the May or June 2026 value, or if a third non-zero snapshot in the same window produces $|1 - \gamma_{\text{July}}/\gamma_{\text{June}}| > 2$. Because the inversion went from strongly above-mean to strongly below-mean (and through zero, which is the canonical “uncoupled” regime), the actionable rule is: future studies must report model snapshot, version date, and repeat the measurement under at least $M \geq 2$ snapshots separated by ≥ 2 weeks; this rule is operationalized as a falsifiable design constraint (drift envelope ≤ 0.5 in γ across snapshots) and is refuted if any single-snapshot γ in a future study is reused more than 4 weeks after measurement without re-validation.

Implication (operationalized as a falsifiable design rule): coupling benchmarks must be versioned and re-validated, with re-validation required at intervals $\Delta t \leq 4$ weeks. The claim “a measurement valid today may be invalid next month” is operationalized as: a single-snapshot γ measurement loses operational validity if its magnitude drifts by $|\Delta\gamma| > 0.5$ within a 4-week window, with the specific evidentiary lower bound $|\Delta\gamma| \geq 1.176$ established by the May–June 2026 GPT-4o comparison ($N=8$ per snapshot). This is refuted if the July 2026 GPT-4o snapshot lies within $|\gamma - 0.000| \leq 0.2$ (i.e., June 2026 holds), or if $|\Delta\gamma| \leq 0.2$ between any two consecutive monthly snapshots over a 3-month follow-up.

7 Implications

7.1 For Experimental Design

1. **Use within-condition designs.** Between-condition comparisons confound evaluator identity with protocol parameters. Within-condition designs varying α_{eval} provide cleaner causal evidence.
2. **Use $N \geq 20$ seeds.** Our simulation shows that at $N=5$, the false-positive probability for a decision-boundary conclusion is 13%. At $N=30$, this drops below 2%.
3. **Use $T \geq 30$ rounds.** Strategy distributions require $T \geq 30$ to approach steady state. Pilot studies with $T < 15$ will observe near-uniform weights regardless of evaluator strength.
4. **Report γ , H , and CV jointly.** The impossibility triangle implies that reporting only one metric provides an incomplete picture.

Bound-guided sample allocation (decision case). The conditional precision floor $\text{CV}(N) \geq 1/(\gamma\sqrt{Ns(\gamma)(1-s(\gamma))})$ can be inverted to set confirmatory sample sizes before collection (script `evidence/scripts/p2_bound_decision_case_20260808.py`; output `analyses/p2_bound_decision_case_20260808.json`). Targeting $\text{CV}=0.30$ (the boundary of the empty favorable regime) yields the required seed counts: for the high-coupling conditions ($\gamma \in [0.72, 1.18]$) the bound demands $N \geq 45\text{--}96$, for the intermediate conditions ($\gamma \approx 0.19\text{--}0.99$) $N \geq 53\text{--}1,289$, and for the self-evaluation conditions ($\gamma \leq 0.19$) $N \geq 1,289\text{--}40,333$. A naive “ $n=5$ seeds suffice” rule is therefore rejected for all eleven conditions at this target; even the most favorable condition requires $N \geq 44$. This changes an allocation decision: an evaluator planner who uses the bound pre-registers N from the floor rather than from an experience rule, and for low- γ candidates either budgets the larger N or demotes them to exploratory status. Because Proposition 2 shows the floor can be non-tight (the proxy variance may far exceed the Bernoulli floor), the inverted N is a conservative reading of the required sample size: in the failure regime the true requirement is at least as large, so the inversion should be treated as the planning minimum and verified empirically before confirmatory use. This only strengthens the rejection of the $n=5$ heuristic, whose shortfall the bound already exposes. We confirm the allocation gain with a ranking simulation on the Chatbot Arena archive (20 models; script `evidence/scripts/p2_bound_vs_heuristic_20260808.py`): recovering the full-sample top-5 model set, the $n=100$ per-model heuristic achieves 76.9% set-overlap, while the bound-guided allocation (per-model n^* from the inverted Bernoulli SE at $\text{SE}=0.03$) achieves 88.1% (+11.2 percentage points) at $2.72\times$ the observations; exact top-5 order recovery rises from 0.7% to 2.7%. The bound therefore trades a predictable observation budget for materially more stable ranking decisions, whereas the heuristic under-samples the low-variance leaders and over-samples the high-variance tail.

7.2 For Evaluator Selection

The bimodal frontier (clusters at $\gamma \in [0, 0.2]$ with high CV, or $\gamma \in [0.7, 1.1]$ with low CV; the favorable regime $\{\gamma < 0.2, \text{CV} < 0.3\}$ is empty across 11 conditions) implies a binary choice:

- **Low-coupling evaluators** (e.g., self-evaluation): Unbiased but unreliable. Require $N \geq 30$ seeds for stable conclusions.
- **High-coupling evaluators** (e.g., cross-model): Biased but reliable. Produce stable measurements at $N \geq 5$ seeds, but the bias must be acknowledged.

There is no “best” evaluator : the choice depends on whether the application prioritizes unbiasedness or reliability.

8 Discussion

What we know. A positive product floor follows under the explicit logistic model and entropy-floor assumption. The reproduced empirical grid shows a negative γ - H association and no point in the selected favorable region. It does not establish that a different evaluator, protocol, estimator, or threshold cannot populate that region. Chatbot Arena results are observational associations with different estimands.

What we don’t know. Whether the triangle extends beyond TTRL to other adaptation mechanisms (e.g., prompt tuning, RLHF). Whether the favorable regime $\{\gamma < 0.2, CV < 0.3\}$ can be populated by multi-model ensembles or calibration interventions. A complementary perspective on what we don’t know comes from recent bounded-memory testbeds for long-horizon agents [Cheng et al. \(2026\)](#): while our impossibility triangle constrains the measurement side, the memory-contract view of [Cheng et al. \(2026\)](#) constrains the agent side, and their interaction (does an explicit memory contract help populate the favorable regime?) is unexplored. Self-evolving rubrics for skill use [Zhu et al. \(2026\)](#) and explicit agent abstention signals [Luo et al. \(2026\)](#) are two additional axes that could shift the triangle’s empirical location and warrant investigation.

Limitations. The theoretical results assume binary observations and multiplicative reweighting. The empirical validation uses a limited set of LLM families. The GPT-4o version drift finding is based on two snapshots; continuous monitoring is needed to characterize drift dynamics. The reported $CV(N=5)$ values are seed-subset dependent (five-seed subsample ranges: DS $\times$ Qwen $[0.587, 2.236]$; Ablation max $[0.145, 0.511]$), and the empirical γ proxy tracks JSD nearly perfectly (Spearman 0.998), so the theory-to-estimator gap is quantitative rather than merely qualitative.

9 Conclusion

We provide a conditional precision and product bound for a stylized TTRL observation model and an explicit audit showing where the current empirical estimators depart from that model. The observed empty region and correlations remain descriptive until all condition rows, alternative metrics, threshold sensitivity, and held-out tests are available. This narrower contribution is preferable to a universal impossibility claim unsupported by the current observation model.

Broader Impact

The impossibility triangle has implications for the safety of LLM-based evaluation systems, with quantitative magnitudes drawn from the 11-conditions dataset: systems optimized for measurement reliability (high γ , e.g., Ablation max $\gamma = 1.038$, $CV(N=5) = 0.157$) produce stable measurements (low CV) but introduce measurable evaluator preference bias; systems optimized for unbiasedness (low γ , e.g., DS Self-Eval $\gamma = 0.033$, $CV(N=5) = 2.42$) produce unbiased estimates but with substantially higher measurement noise (CV inflated by $15.4\times$ relative to Ablation max). The “may” framing of these safety claims is operationalized as: a high-coupling system introduces systematic bias if the induced ΔH (entropy drop from baseline) exceeds 0.05 at $T=30$; a low-coupling system produces unreliable measurements if $CV \geq 1.0$ at $N=5$. In high-stakes applications where evaluation drives model selection, the tradeoff must be explicitly managed, with a falsifiable rule: model selection decisions that change the evaluator must be re-validated against the new evaluator’s γ profile, and the rule is refuted if a re-evaluation under a new evaluator produces $|\Delta \text{accuracy}| \leq 0.01$ across a fixed benchmark (which would imply the bias has no measurable impact on selection).

Reproducibility

All experiments use official API endpoints with fixed model versions. The standardized protocol, reference snapshot, and analysis code are released under the MIT License at <https://anonymous.4open.science/r/paper2-anon-638E/>. The release contains:

- `analysis/impossibility_triangle.py`, which recomputes the γ_{HCV} product for the 11 evaluator-agent conditions and classifies each condition using Table 5;
- `analysis/ttrl_simulator.py`, a deterministic TTRL simulator with $K=11$ strategies and configurable α_{eval} ;
- `analysis/cv_estimator.py`, which evaluates the Cramer-Rao and tight continuous-weight bounds; and
- `analysis/arena_loader.py`, which reproduces the Chatbot Arena entropy-decisiveness analysis.

Bundled offline data include `data/chatbot_arena_sample.jsonl` (200 rows), `data/convergence_pilot.jsonl` (28 rows), and `data/triangle_11_conditions.csv`. All scripts use a fixed default seed of 42 and expose a `-seed` option. Raw per-seed observations for the 11-condition summary (approximately 3 GB of API logs) are available from the corresponding author. The short-window robustness claim is refuted if any 30-day sub-window yields $|r| < 0.5$ with $p \geq 0.05$, or if a 50% model-retention resample produces a 95% confidence interval spanning zero.

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